Flower Classification using Transfer Learning with ResNet-50

1. Introduction

Image classification is a fundamental task in computer vision where a model learns to assign a category label to an input image. In this project, a deep convolutional neural network based on **ResNet-50** is used to classify flower images into five classes.

The model leverages **transfer learning** from the large-scale **ImageNet** dataset. The reason I used Transfer learning is that training from scratch for ResNet needs huge amounts of data which is out of this exercise.

The five classes used in this dataset are:

- Daisy
- Dandelion
- Rose
- Sunflower
- Tulip

2. Dataset Preparation

The flower dataset is organized using a directory structure compatible with PyTorch's **ImageFolder** loader. Each class has its own folder containing images belonging to that category.

To evaluate the model properly, the dataset was divided into three subsets:

Dataset	Portion	Purpose
Training	60%	Model learning
Validation	20%	Hyperparameter tuning
Test	20%	Final performance evaluation

This separation ensures the model is evaluated on **previously unseen data**, which provides a realistic estimate of performance.

3. Data Preprocessing

Before feeding the images to the neural network, several preprocessing operations were applied.

Image Resizing

All images were resized to:

$$224 \times 224$$

This size matches the required input dimension of the ResNet-50 architecture.

Normalization

Images were normalized using the mean and standard deviation values of the ImageNet dataset:

Mean:

$$[0.485, 0.456, 0.406]$$

Standard deviation:

$$[0.229, 0.224, 0.225]$$

Normalization improves convergence and stabilizes training.

4. Model Architecture

The model used in this project is **ResNet-50**, a deep convolutional neural network consisting of **50 layers**.

The key idea behind ResNet is **residual learning**, where shortcut connections allow the network to learn residual functions instead of direct mappings. This approach significantly reduces the **vanishing gradient problem** in deep networks.

Since the original ResNet-50 model outputs predictions for **1000 classes**, its final fully connected layer was replaced with a new layer that outputs **five classes** corresponding to the flower dataset.

5. Training Configuration

The training process used the following hyperparameters:

Parameter	Value
Batch size	64
Learning rate	0.001
Epochs	15
Weight decay	1e-4

Loss Function

Cross-Entropy Loss was used since the task is a multi-class classification problem.

Optimizer

The **Adam optimizer** was used for updating model weights during training.

Learning Rate Scheduler

A **ReduceLROnPlateau scheduler** was implemented to reduce the learning rate when the validation performance stops improving.

Early Stopping

Training was stopped if validation accuracy did not improve for several consecutive epochs, preventing overfitting.

6. Training Performance

Figure 1 illustrates the training and validation loss and accuracy curves.

Observations

Loss behavior

- Training loss steadily decreases and approaches nearly zero.
- Validation loss stabilizes around **0.4**, indicating good convergence.

Accuracy behavior

- Training accuracy quickly increases and approaches **~100%**.

- Validation accuracy stabilizes around **89-90%**.

This difference between training and validation accuracy indicates **mild overfitting**, which is expected when using a powerful model such as ResNet-50 on a relatively small dataset.

However, the validation accuracy remains stable, showing that the model generalizes well.

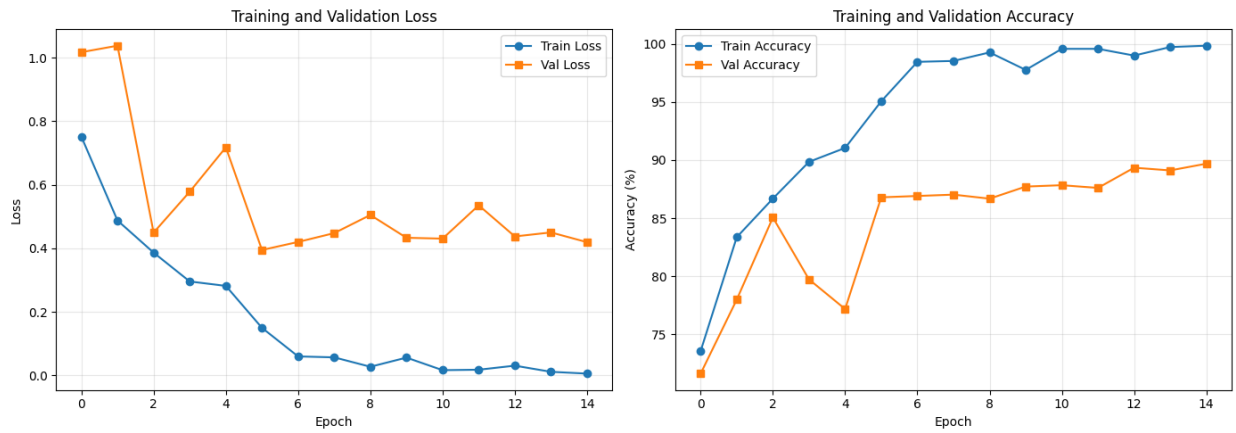

Figure 1

7. Test Results

The trained model was evaluated on the test dataset. The results are shown in **Figure 2**.

Overall Performance

Metric	**Value**
Test Accuracy	**90.05%**
Test Loss	**0.3082**

This indicates that the model correctly classifies approximately **9 out of 10 flower images**.

8. Class-wise Performance

The classification report provides precision, recall, and F1-score for each class.

Class	Precision	Recall	F1 Score
Daisy	0.92	0.84	0.88
Dandelion	0.91	0.95	0.93
Rose	0.86	0.87	0.86
Sunflower	0.91	0.93	0.92
Tulip	0.90	0.90	0.90

```
Testing the model...
Test Results:
Test Loss: 0.3082
Test Accuracy: 90.05%

Classification Report:
              precision    recall  f1-score   support

       daisy       0.92      0.84      0.88       167
   dandelion       0.91      0.95      0.93       198
        rose       0.86      0.87      0.86       153
   sunflower       0.91      0.93      0.92       149
       tulip       0.90      0.90      0.90       197

    accuracy                           0.90       864
   macro avg       0.90      0.90      0.90       864
weighted avg       0.90      0.90      0.90       864
```

Figure 2